

Business Sales Performance Analytics

Revenue Trends, Product Performance & Regional Insights

Prepared by Salathiso Ntsele | Future Interns: Data Science & Analytics Track | Task 1 | Dataset: Sample Superstore (2014–2017, 9,994 transactions)

\$2.30M Total Revenue | **\$286K** Total Profit | **12.5%** Profit Margin Value | **5,009** Orders | **\$458.61** Avg. Order Value

Executive Summary

Over four years (2014–2017), the business generated \$2.30M in revenue across 5,009 orders from 793 customers, at a 12.5% overall profit margin. Revenue grew every year except a slight dip in 2015, with 2017 revenue up 20.4% year-on-year on the back of a strong Q4. Technology and Office Supplies are the most profitable categories, while Furniture drags overall margin down to just 2.5%, largely due to heavy discounting on Tables and Bookcases. The West and East regions are the strongest performers; Central lags on margin, and four states (Texas, Ohio, Pennsylvania, Illinois) are net loss-makers despite solid sales volumes. The clearest lever for margin improvement is discount discipline: orders discounted above 20% are unprofitable as a group, wiping out roughly \$135K in profit.

1. Revenue Trends

Monthly revenue shows a clear seasonal pattern: sales are consistently strongest in the final quarter of each year (November–December), driven by holiday-period demand, and weakest in the first quarter. This rhythm repeats across all four years and should anchor inventory and cash-flow planning.

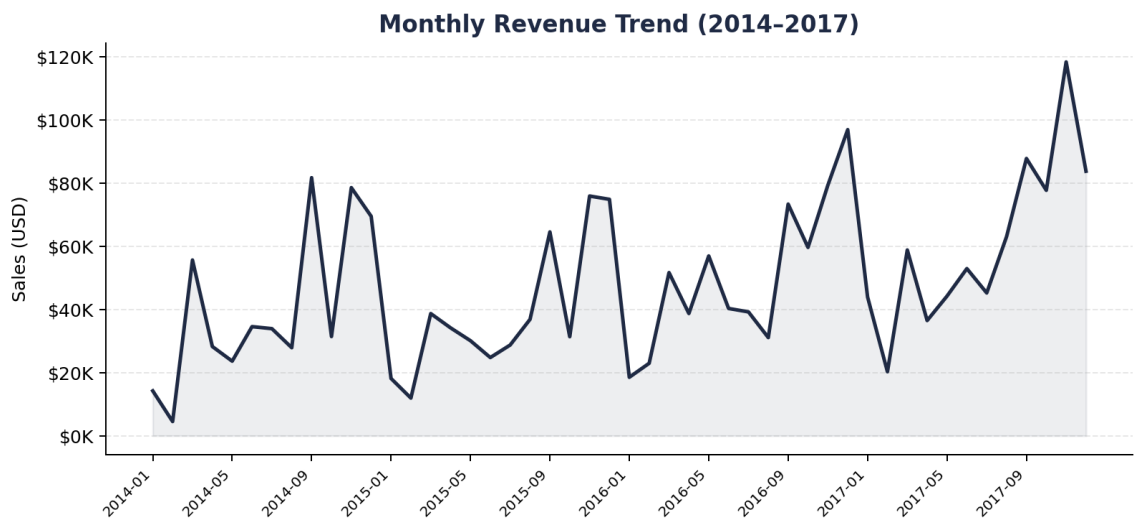


Figure 1. Monthly revenue, January 2014 – December 2017.

Year-over-year, revenue dipped slightly in 2015 (\$470.5K, -2.8%) before accelerating to \$609.2K in 2016 and \$733.2K in 2017, a 20.4% jump. Profit followed a similar upward path, though it grew more slowly than revenue in 2016, indicating rising discount pressure that year before margins recovered in 2017.

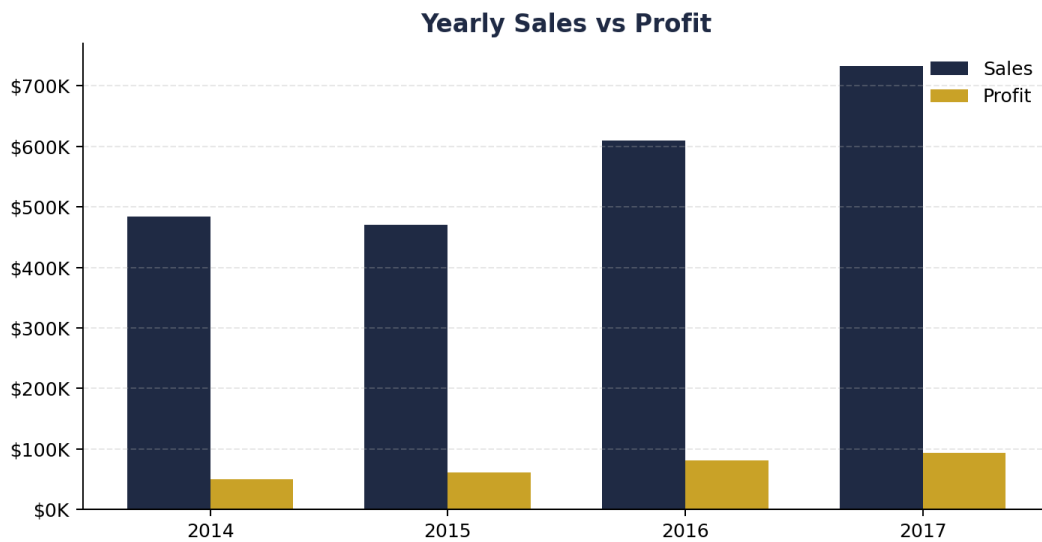


Figure 2. Annual sales versus profit, 2014–2017.

2. Top-Selling Products

Revenue is concentrated in a small number of high-ticket items. The Canon imageCLASS 2200 copier alone generated \$61.6K, more than double the next best-seller, followed by a binding machine, a Cisco videoconferencing unit, and premium office chairs. These are low-volume, high-value items rather than everyday staples, suggesting the business's revenue depends heavily on B2B / capital-equipment style purchases.

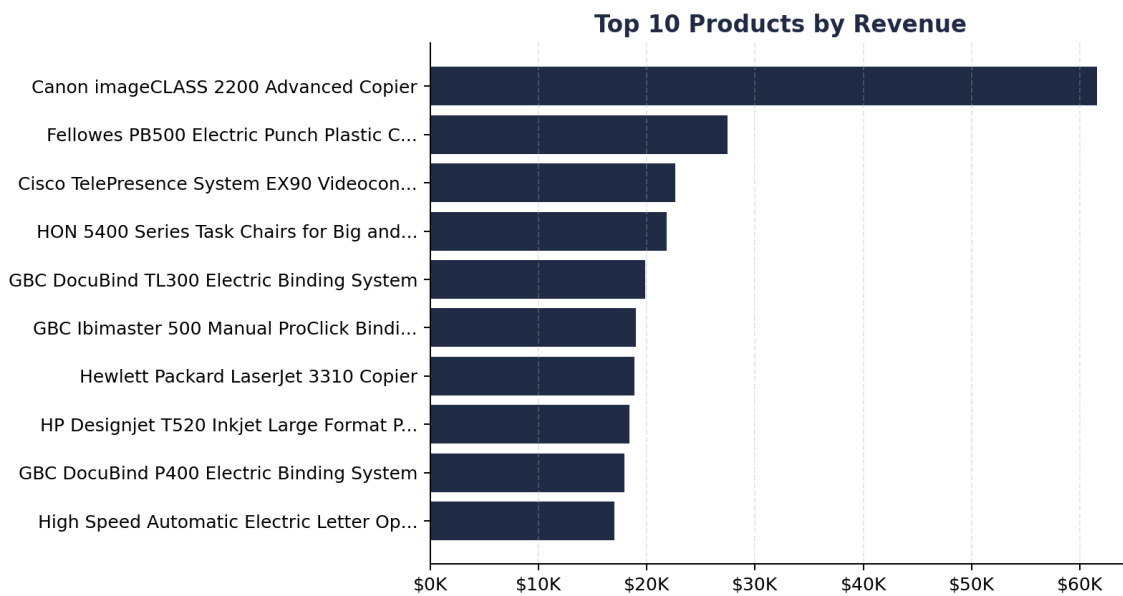


Figure 3. Top 10 products ranked by total revenue.

Product	Revenue
Canon imageCLASS 2200 Advanced Copier	\$61,600
Fellowes PB500 Electric Punch Binding Machine	\$27,453
Cisco TelePresence System EX90	\$22,638
HON 5400 Series Task Chairs (Big & Tall)	\$21,871

Product	Revenue
GBC DocuBind TL300 Electric Binding System	\$19,823

3. Category & Sub-Category Performance

The three product categories generate broadly similar revenue, but profitability differs sharply. Technology (\$836K sales, 17.4% margin) and Office Supplies (\$719K sales, 17.0% margin) are healthy. Furniture (\$742K sales) converts to just 2.5% margin, nearly all of it eroded by two sub-categories.

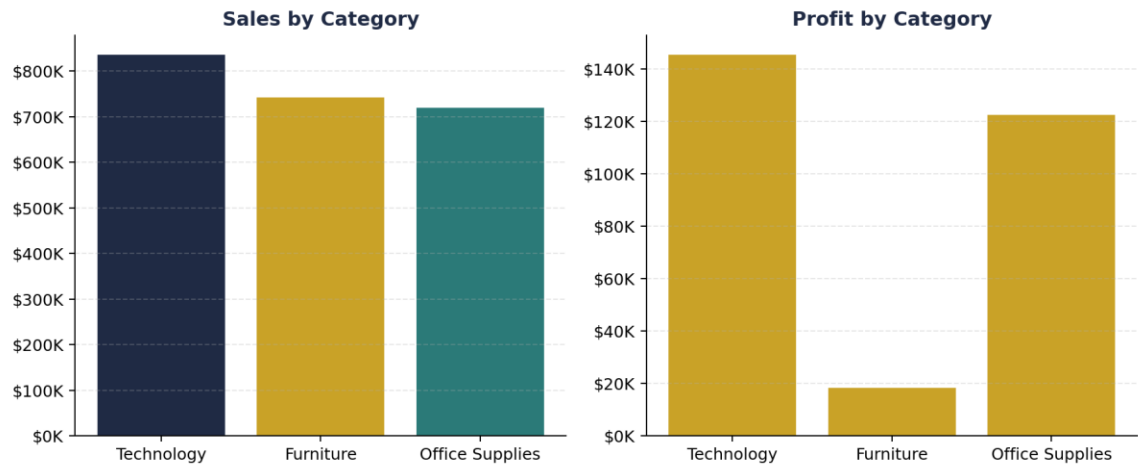


Figure 4. Sales and profit by category.

Breaking Furniture down further shows the source of the problem: **Tables lose \$17.7K** (-8.6% margin) and **Bookcases lose \$3.5K** (-3.0% margin), while Chairs and Furnishings remain profitable. At the other end, Copiers (37.2% margin), Paper (43.4%), Labels (44.4%) and Envelopes (42.3%) are the most profitable sub-categories per rand of revenue.

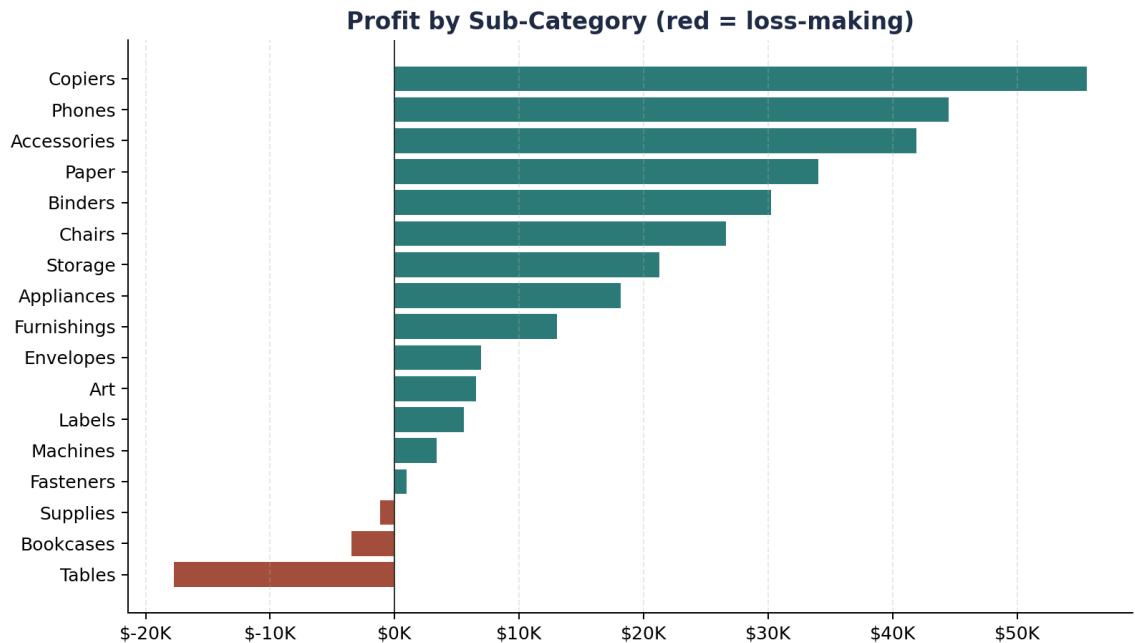


Figure 5. Profit by sub-category: Tables and Bookcases are the only loss-makers.

4. Regional Performance

The West region leads on both revenue (\$725K) and profit (\$108K, 14.9% margin), followed by the East (\$679K sales, 13.5% margin). Central generates solid sales (\$501K) but the weakest margin of any region (7.9%), and the South, while smallest in revenue, holds a respectable 11.9% margin.

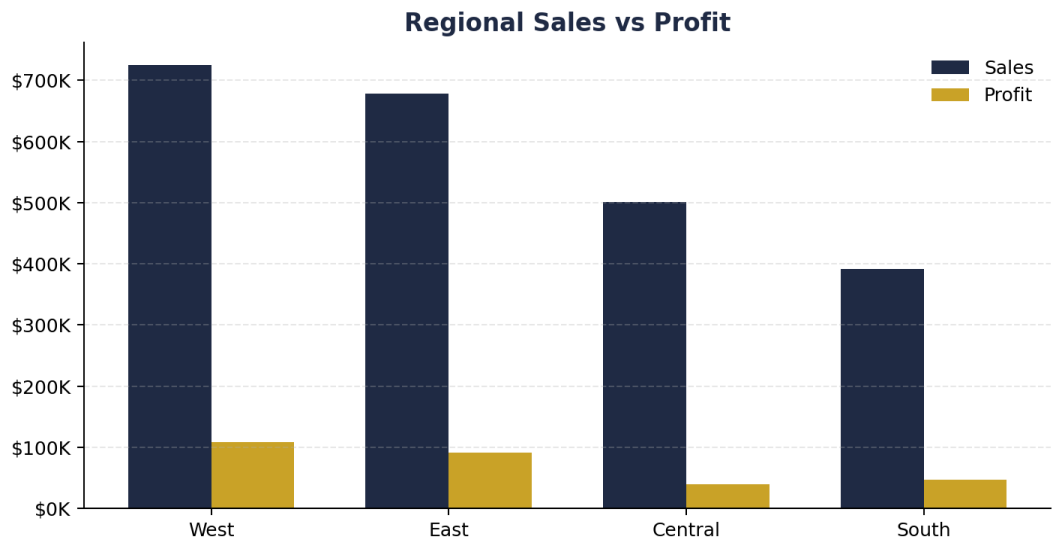


Figure 6. Sales versus profit by region.

At the state level, California (\$457.7K) and New York (\$310.9K) are the two largest and most profitable markets. However, four states are net loss-makers despite meaningful sales volume: **Texas (-\$25.7K profit on \$170.2K sales)**, Ohio (-\$17.0K), Pennsylvania (-\$15.6K), and Illinois (-\$12.6K), together representing roughly \$71K in lost profit that a healthier discount policy could largely recover.

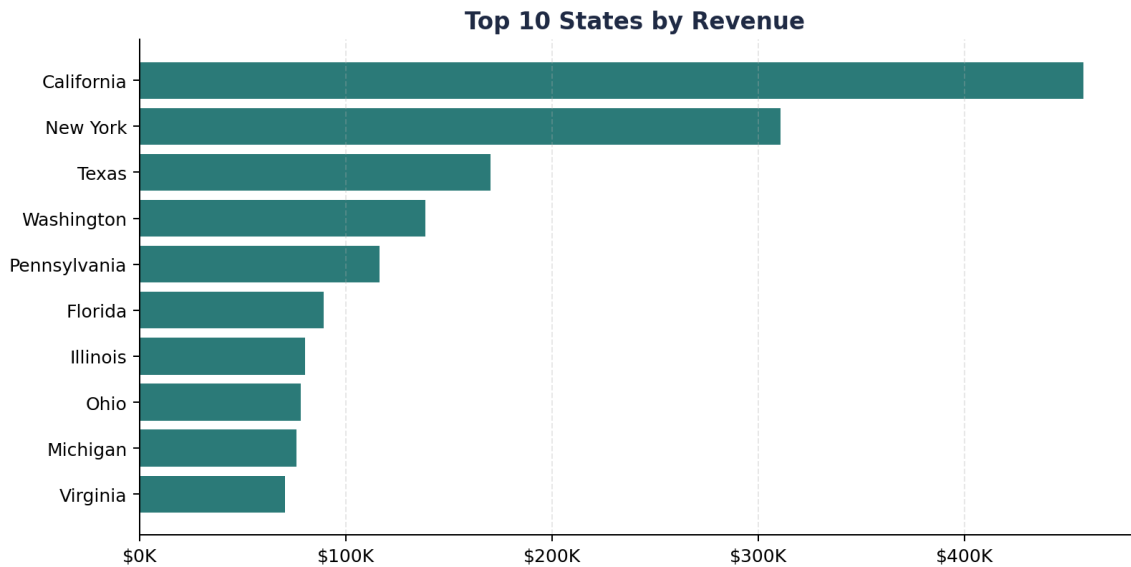


Figure 7. Top 10 states by revenue.

5. What Is Driving the Margin Loss? Discounting

Discounting is the single clearest driver of unprofitability in this dataset. Orders with no discount or up to 20% off remain solidly profitable. Once the discount exceeds 20%, orders flip to a net loss as a group, the 20-40% discount band alone erased \$35.8K in profit, and discounts above 40% erased a further \$99.6K, despite generating only \$128K in sales. In other words, roughly \$135K of profit, nearly half of total profit earned, is being given away through deep discounting on a relatively small slice of orders.

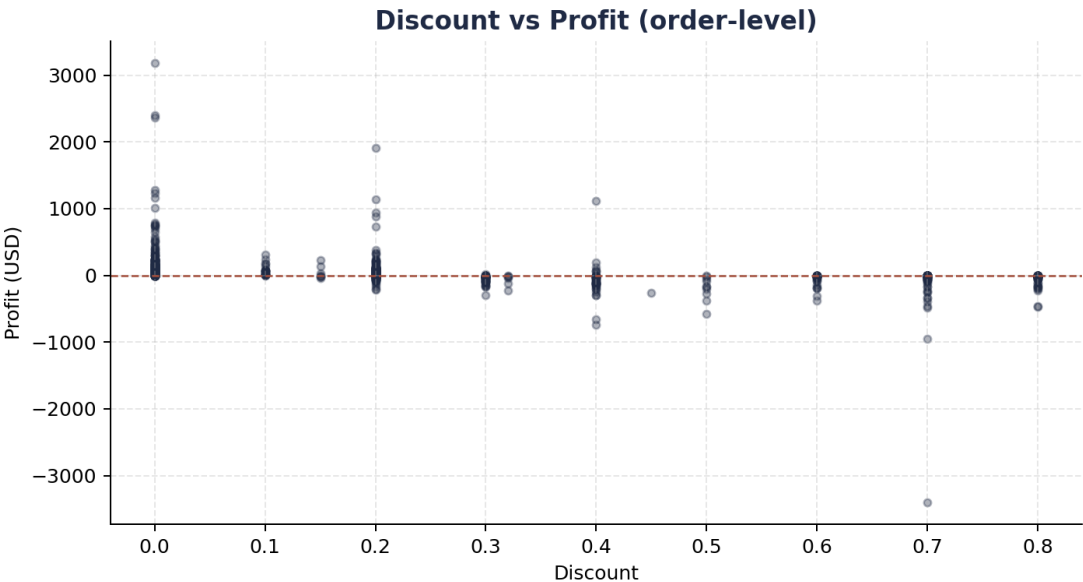


Figure 8. Discount level versus order profit — losses cluster above ~20% discount.

Discount Band	Sales	Profit	Orders
0% (no discount)	\$1,087,908	\$320,988	4,798
1–20%	\$846,522	\$100,785	3,803
21–40%	\$234,138	-\$35,817	460
41%+	\$128,632	-\$99,559	933

6. Customer Segments

Consumer is the largest segment by revenue (\$1.16M) but has the lowest margin (11.5%). Home Office is the smallest segment (\$430K) but the most profitable per rand of sales (14.0% margin), followed by Corporate (13.0%). This suggests Home Office and Corporate customers are comparatively higher-value and may merit targeted retention or upsell attention.

7. Actionable Recommendations

1. Cap discounts above 20%

Introduce an approval threshold for discounts beyond 20%, especially on Tables and Bookcases. This single band accounts for roughly \$135K in lost profit and is the fastest lever available.

2. Re-price or redesign the Tables line

Tables lose 8.6% margin overall, the worst of any sub-category. Review supplier cost, freight, and list pricing before continuing to discount this line at current rates.

3. Fix the Texas, Ohio, Pennsylvania and Illinois markets

These four states are net loss-makers despite \$485K in combined sales. Investigate local discounting practices and shipping costs; even bringing them to break-even would add ~\$71K in profit.

4. Double down on Technology and high-margin Office Supplies

Copiers, Paper, Labels and Envelopes carry the strongest margins (37-44%). Prioritise marketing spend and stock availability for these lines, particularly ahead of the Q4 seasonal peak.

5. Plan inventory and cash flow around the Q4 seasonal spike

Revenue consistently peaks in November-December every year. Build purchasing and staffing plans around this predictable cycle rather than treating each year's spike as a surprise.

6. Protect and grow Home Office / Corporate relationships

These segments carry the best margins per rand of sales. A loyalty or account-management push here is likely to be more profit-accretive than acquiring more Consumer-segment volume.

Methodology & Data Notes

Analysis performed in Python (pandas, matplotlib) on the Sample Superstore dataset, 9,994 order-line transactions, January 2014 to December 2017, with no missing values or duplicate rows. Sales and profit figures are aggregated at the order-line level; 'margin' refers to profit as a percentage of sales. Currency figures are as recorded in the source dataset (USD).